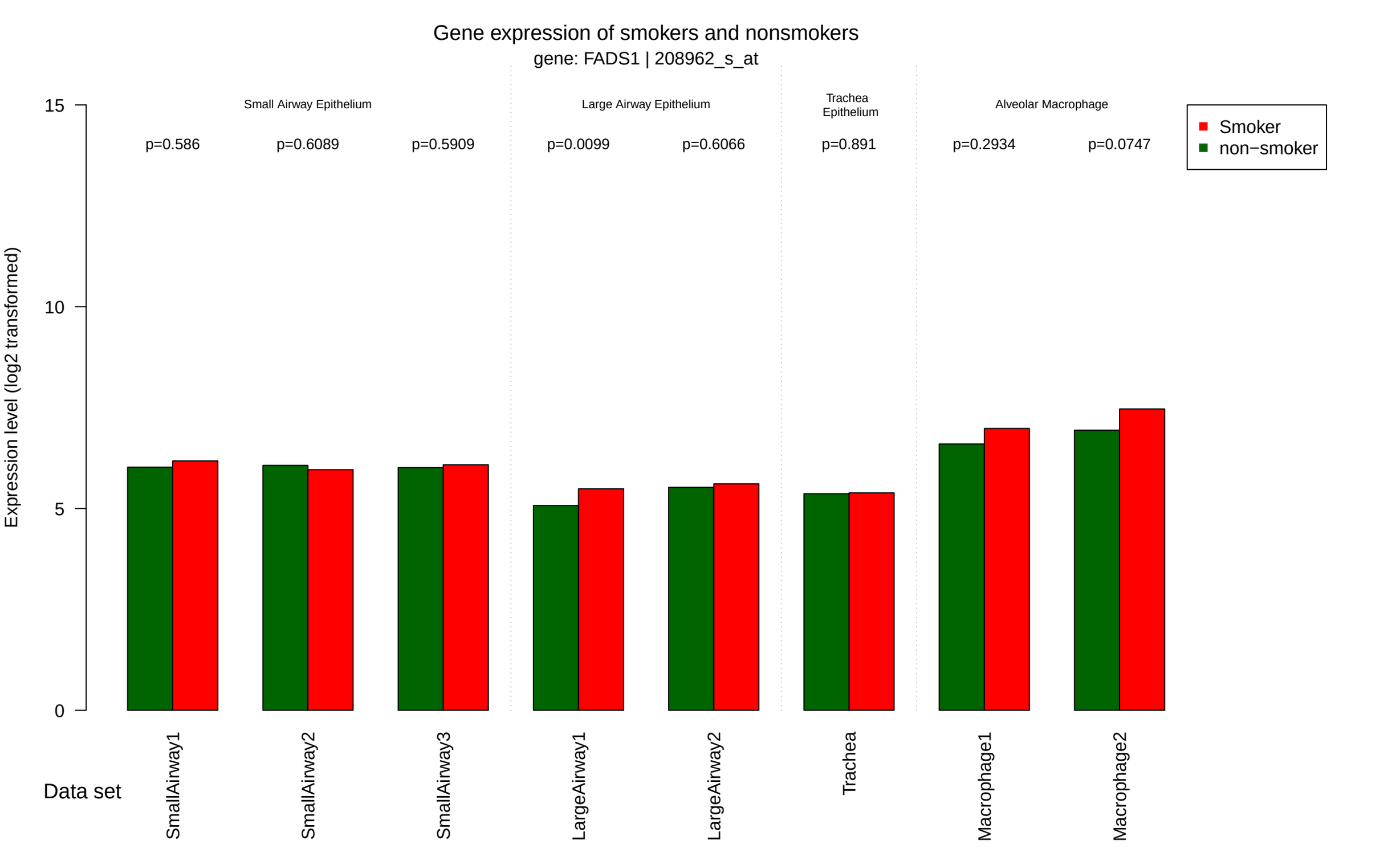
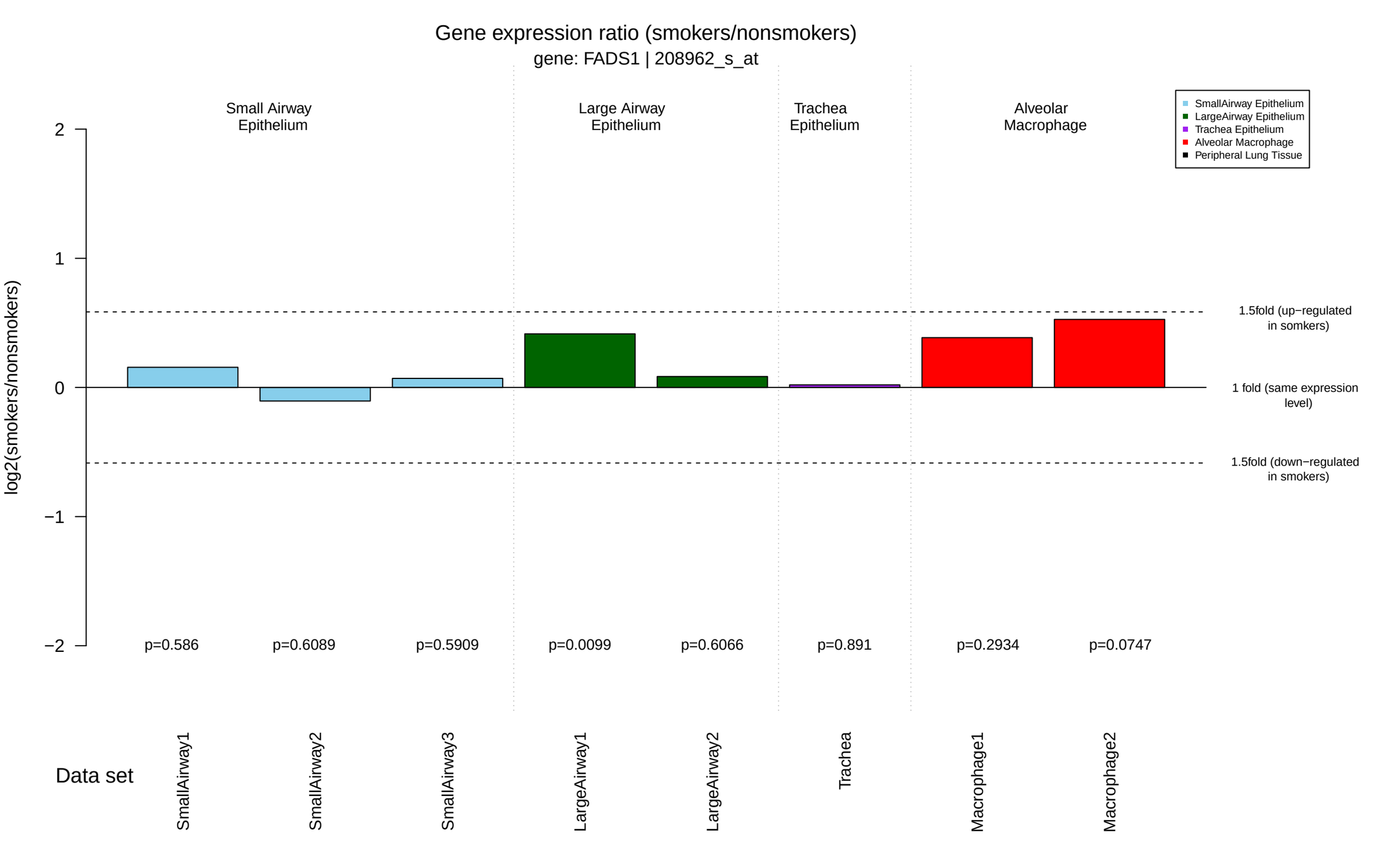
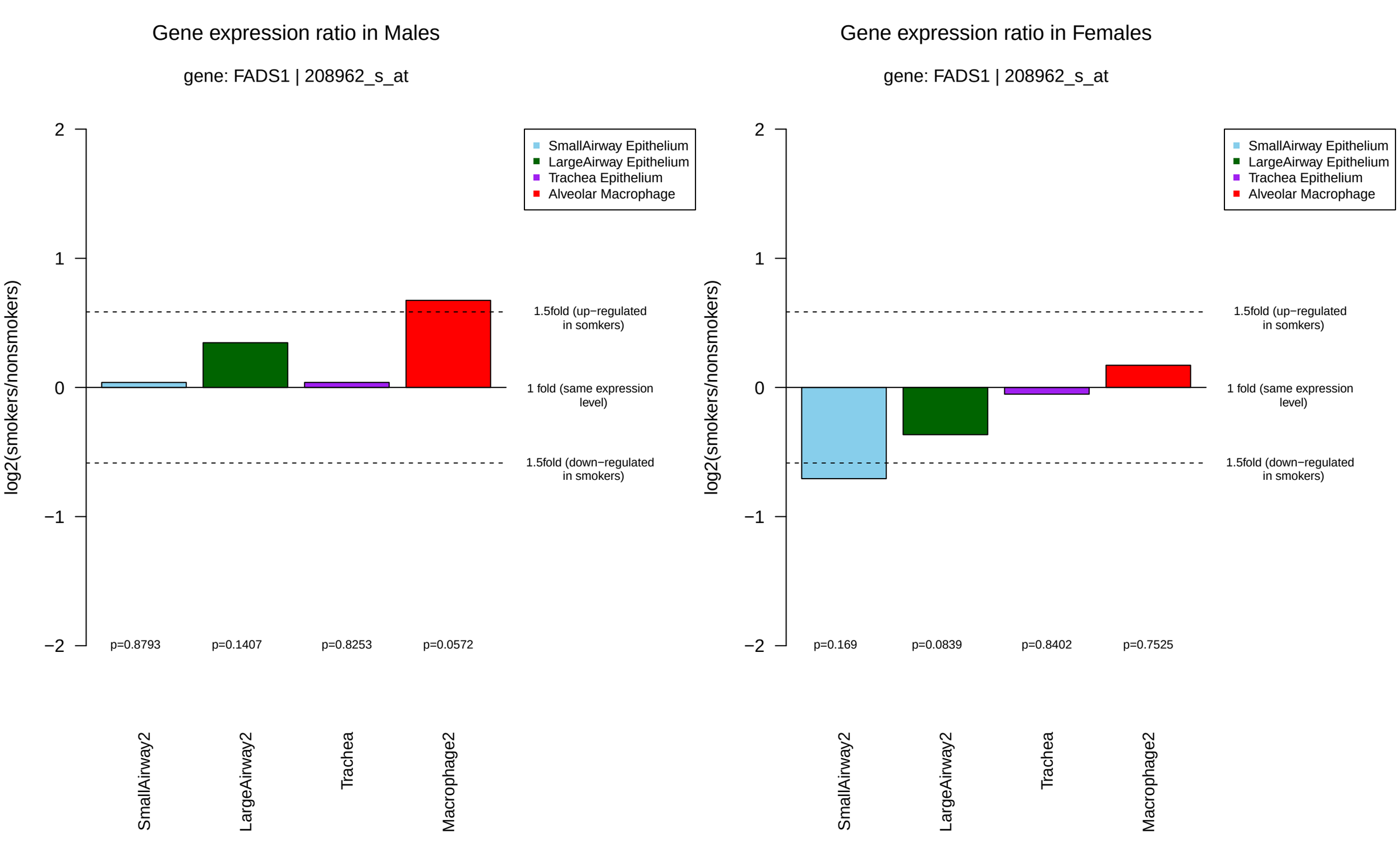
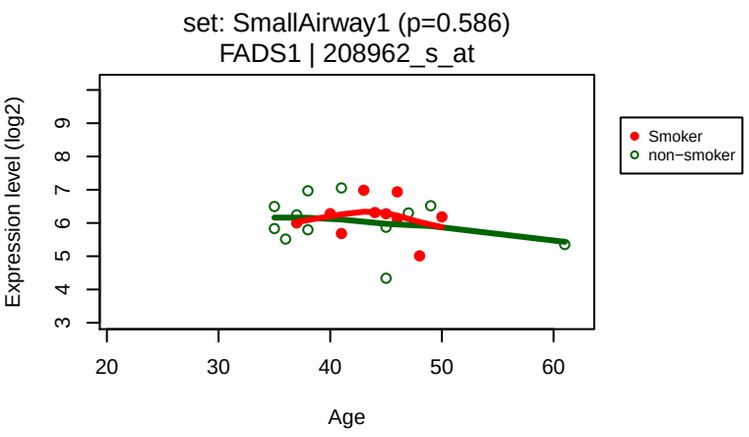
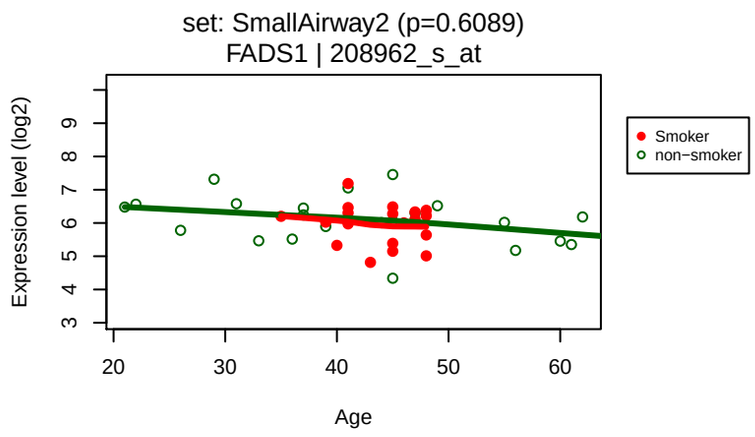
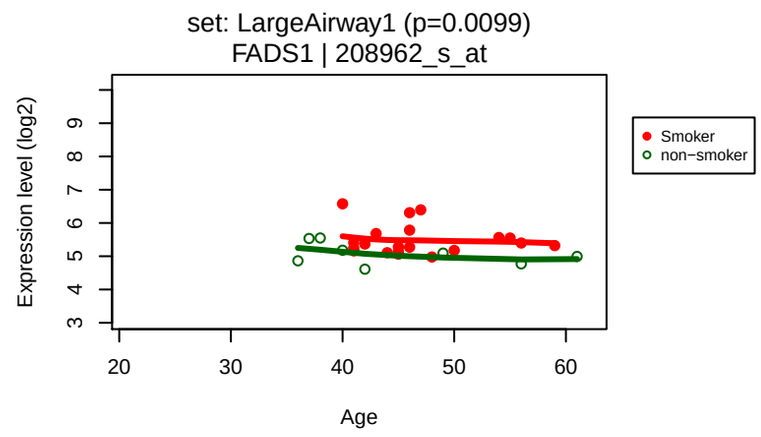

Gene expression of smokers and nonsmokers
gene: FADS1 | 208962_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.586
p=0.6089
p=0.5909
p=0.0099
p=0.6066
p=0.891
p=0.2934
p=0.0747
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

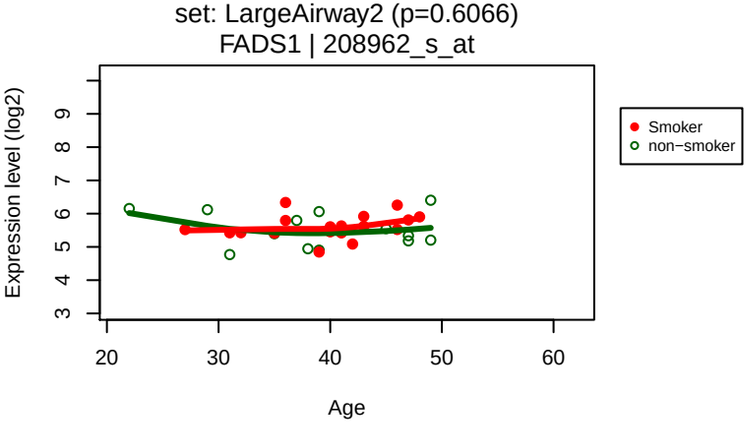

Gene expression ratio (smokers/nonsmokers)
gene: FADS1 | 208962_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.586
p=0.6089
p=0.5909
p=0.0099
p=0.6066
p=0.891
p=0.2934
p=0.0747
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

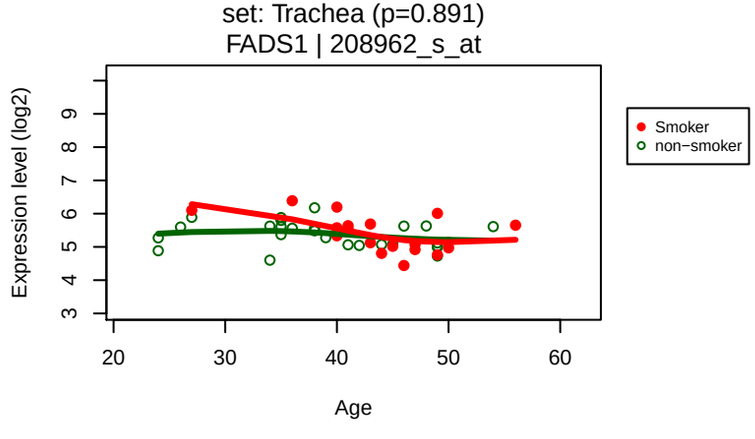

Gene expression ratio in Males
gene: FADS1 | 208962_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8793
p=0.1407
p=0.8253
p=0.0572
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FADS1 | 208962_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.169
p=0.0839
p=0.8402
p=0.7525
SmallAirway2
LargeAirway2
Trachea
Macrophage2

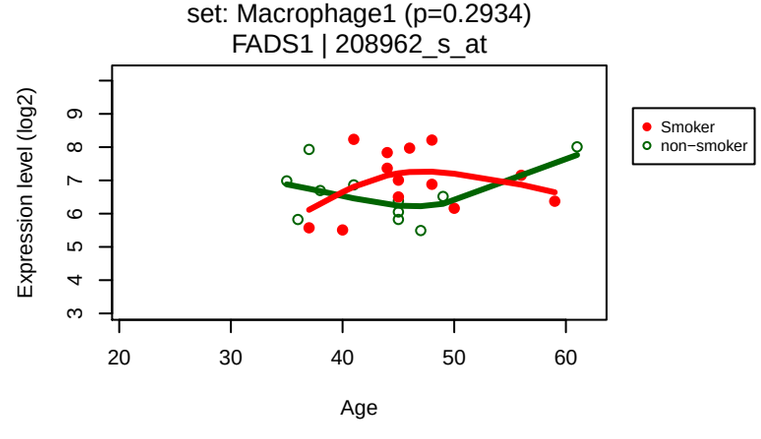

set: SmallAirway1 (p=0.586)
FADS1 | 208962_s_at
Expression level (log2)
Age
Smoker
non−smoker

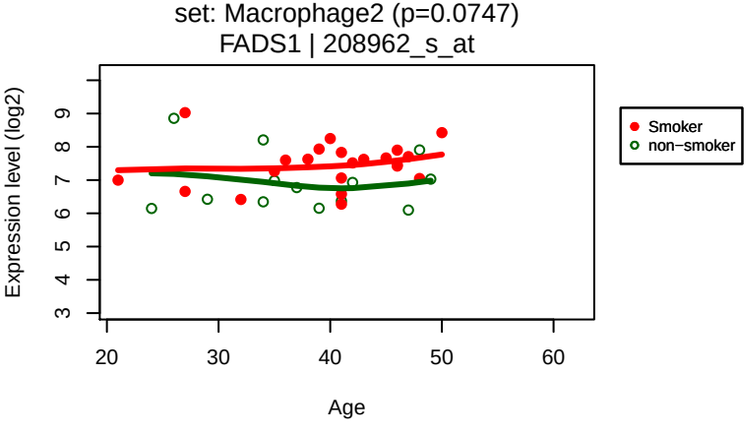

set: SmallAirway2 (p=0.6089)
FADS1 | 208962_s_at
Expression level (log2)
Age
Smoker
non−smoker

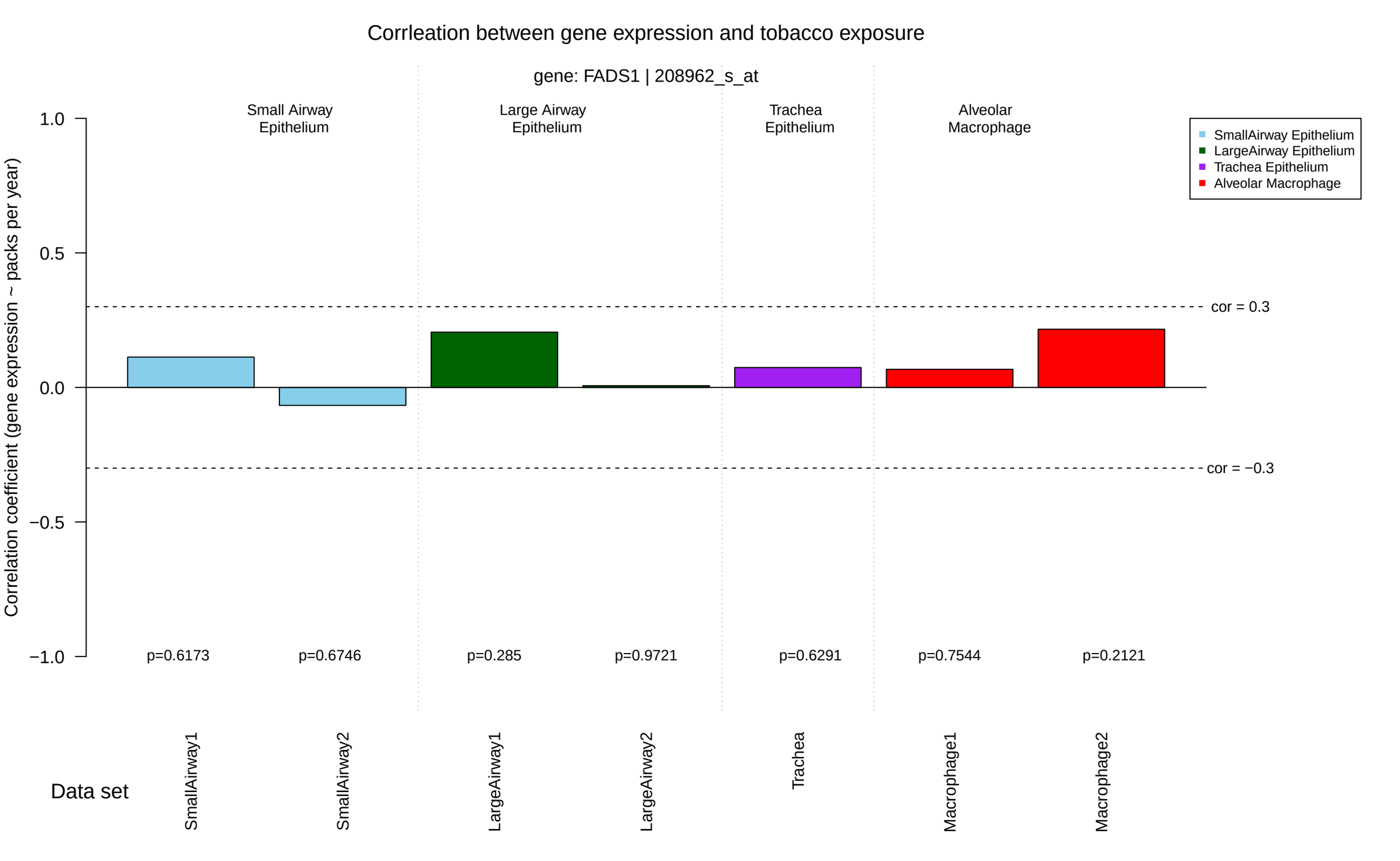

set: LargeAirway1 (p=0.0099)
FADS1 | 208962_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6066)
FADS1 | 208962_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.891)
FADS1 | 208962_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2934)
FADS1 | 208962_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0747)
FADS1 | 208962_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FADS1 | 208962_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6173
p=0.6746
p=0.285
p=0.9721
p=0.6291
p=0.7544
p=0.2121
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2